

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SHORT ANSWER TEST (30 Questions)

Course Code:	ITA0502	Course Title:	COMPUTER VISION
CO Assessed:	CO1 – Imitation (BL3)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	40% of CIA	Total Marks:	50 (10 Questions × 5 Marks)
CO4 Statement:	Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems. (BTL 3)		
Student Name:	Mohd Afnan	Reg. No.:	192421317
Section / Batch:		Date:	

Code of Conduct

I Afnan 192421317 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 6 questions, each carrying 5 marks. Total: 30 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Overview of Computer Vision	Definition, objectives, workflow, and importance of Computer Vision	1	5
Features of Computer Vision	Feature extraction, object recognition, and image understanding	2	5
Levels of Computer Vision and Image Processing	Low-level, mid-level, and high-level image processing operations	3	5
Digital Image Fundamentals	Pixels, image representation, grayscale/color images, and image resolution	4,10	10
Image Sensing and Acquisition	Image sensors, acquisition process, and imaging devices	5	5
Sampling and Quantization	Sampling, quantization, and digital image formation	6	5
Image Formation Models	Perspective projection, illumination, and image formation process	7	5
Applications of Image Processing and Computer Vision	Healthcare, autonomous vehicles, surveillance, agriculture, robotics, and industrial automation	8,9	10
		GRAND TOTAL:	50 Marks

Annexure A – Short Answer Test

- 1. How does Computer Vision enable computers to interpret and understand visual information?**
 - a) What are the main objectives of Computer Vision?
 - b) Mention any two real-world applications of Computer Vision.
- 2. What are the key features of Computer Vision that make it suitable for intelligent image analysis?**
 - a) Define feature extraction.
 - b) How does object recognition improve Computer Vision systems?
- 3. How are the different levels of image processing used in Computer Vision?**
 - a) What is low-level image processing?
 - b) Differentiate between mid-level and high-level image processing.
- 4. How is a digital image represented and what are its fundamental components?**
 - a) Define a pixel.
 - b) How does image resolution affect image quality?
- 5. How does image sensing and acquisition convert a real-world scene into a digital image?**
 - a) What is the role of an image sensor?
 - b) Name any two image acquisition devices.
- 6. How do sampling and quantization contribute to the formation of a digital image?**
 - a) Define sampling.
 - b) What is the purpose of quantization in image processing?
- 7. How does the image formation model describe the process of capturing images?**
 - a) What is perspective projection?
 - b) What is the role of illumination in image formation?
- 8. How are image processing and Computer Vision related, and how do they differ?**
 - a) Define image processing.
 - b) State two differences between image processing and Computer Vision.
- 9. How is Computer Vision applied in solving real-world problems across different domains?**
 - a) Explain one application in healthcare.
 - b) Explain one application in autonomous vehicles.
- 10. How do digital image fundamentals support Computer Vision applications?**
 - a) Why are pixels considered the basic building blocks of digital images?
 - b) How do sampling and image resolution influence image analysis?

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks	
Understanding of Concepts	50%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions		
Presentation	50%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation, lacks depth	Poor or unclear explanation		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Short Answer Test

Mohammed Khan
192421317
2024 Batch

1) a) The main objective of the ~~Computer~~ vision is to interpret and reading the ~~Image~~ and Understanding. and also helps for ~~decision~~ making

b) The main two real world applications of computer vision is face recognition where we used for authentication. and Self Car. driving ~~these~~ are 2 real world applications.

2) a) The feature extraction is define as the process of defining the characteristics of an image like edges, corners, etc...

b) The object recognition system improves the computer vision system by helping to understand the image and telling about the object accurately.

3) In the low level image processing it is deal with basic image operations like with row pixel values.

Low level
Processing

Segmentation and feature extraction

Identifies regions

Act as bridge b/w low level and high level

High level
Processing

Object recognition and scene understanding

Support decision making based on image.

Understand meaning of image.

4) Pixel is the value of the images. computers can't understand images it reads the image in the form of pixel values.

5) The image resolution affects the quality of the image because it makes the image more sharper and bright so the image will affect.

6) as the role of an image sensor is to convert the light into electrical signals.
like, Scanner etc.

7) the acquisition devices are.

* Digital Camera.

* Scanner

a) Sampling is the process of converting the Continuous Image into grid of pixels.
by the purpose of quantization Image process is to fix the Intensity or fix the colors to each pixel.

b) a perspective projection is defines as Projecting a 3D Scene to the 2D Image Plane. object further from camera appears smaller than near by object.
by the role of illumination in image formation is to provide the light source to a captured image. Good lighting improves the image quality.

c) as High Image processing is the process of enhancing, restoring or modifying the digital image which improves image quality and analysis or display.

9. a) Image processing:-

- * It focuses on enhancing images
- * It produces improved images.

Computer vision

- * It focuses on understanding the image.
- * It produces useful info and decisions.

9. a) Computer vision helps to detect the diseases by analysing the medical images like X-rays, CT scan, or MRI scan etc.

b) Computer vision helps to enable self driving cars to detect the road, traffic signals and pedestrians. This helps the vehicle to navigate.

10. a) pixels are smallest unit of a digital image and stores color and brightness. All the digital images are made up of million pixels.

b) higher sampling and image resolution provide more image details, improving accuracy in analysis and object recognition. lower resolution may lose important details.